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Lorenzo Sala (INRAE)

Subject: Application for PhD Position – Compositional Hybrid Digital Twins for Fermented Microbial Ecosystems

Dear Lorenzo Sala,

I am applying for the PhD position on "Compositional Hybrid Digital Twins for Fermented Microbial Ecosystems" with great enthusiasm. With a strong engineering background in Artificial Intelligence, distributed computing, and complex system modeling, I am highly motivated to contribute to the Artemis framework and tackle the unique challenges presented by this project.

Bridging physical and biological systems requires deep expertise in hybrid modeling. In my engineering portfolio, I have extensively utilized Multi-Fidelity Scientific Machine Learning (SciML) and Physics-Informed Neural Networks to model complex thermodynamic and fluid dynamics systems. I am highly motivated to adapt these rigorous hybrid methodologies to model fermented microbial ecosystems, utilizing data-driven AI to complement traditional biochemical models.

I am particularly drawn to this position because it bridges rigorous research with practical industrial challenges within the EDT program. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui